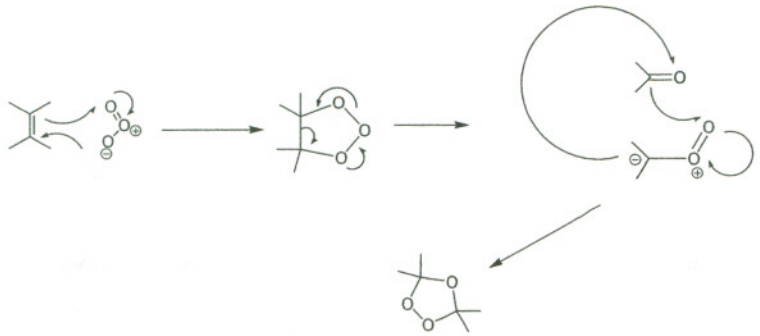
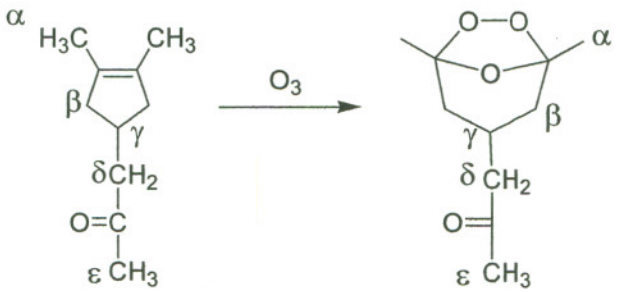
Organic Chemistry Cume - November 11, 2005

1) The currently accepted mechanism of alkene ozonolysis is thought to proceed via three consecutive six-electron pericyclic reactions:

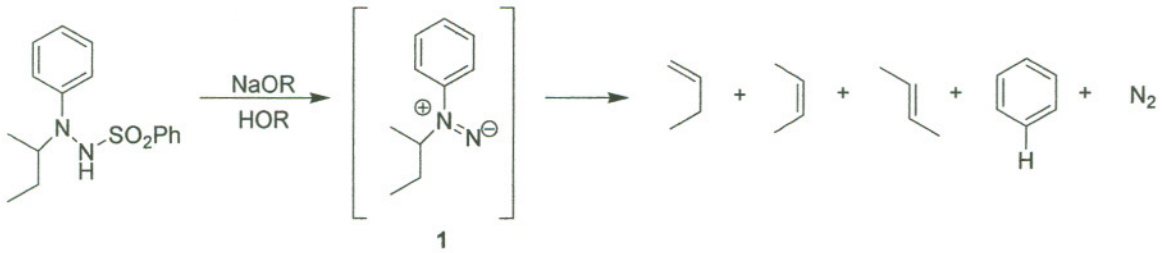


The crucial experiments which have led to this mechanistic hypothesis are almost entirely due to work done by Rudolf Criegee's group. One of the most elegant of these begins with the following reaction:



The reactant alkene was first deuterated fully at the δ and ε positions. Use the Criegee mechanism to predict the percent deuteration of the six α-hydrogens, of the four β-hydrogens, of the one γ-hydrogen, of the two δ-hydrogens, of the three ε-hydrogens in the ozonide product.

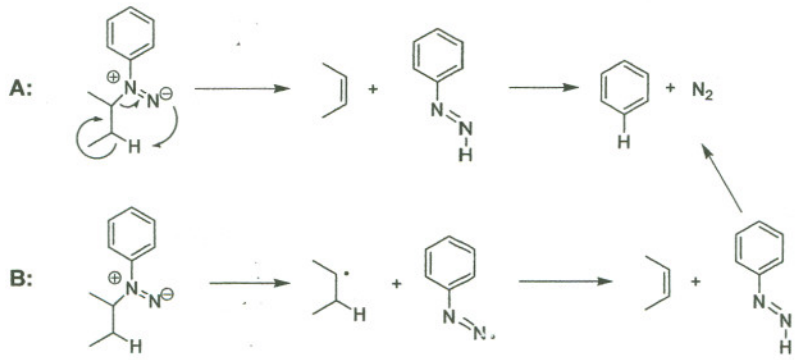
2) Reaction of the sulfonated hydrazine shown below with base provides a mixture of 1- and 2-butenes. The reaction is thought to proceed via the 1,1-diazene intermediate (**1**).



Two mechanisms for the above transformation come to mind:

A: a one step elimination

B: a stepwise process



Suggest an experiment which might distinguish these two mechanisms and be sure to explain what the outcome would be for both mechanisms.